

SASTRA DEEMED UNIVERSITY

(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

February 2023

Course Code: EIE101R01

Course: BASIC ELECTRONICS ENGINEERING

QP No. : U006R-1

Duration: 3 hours

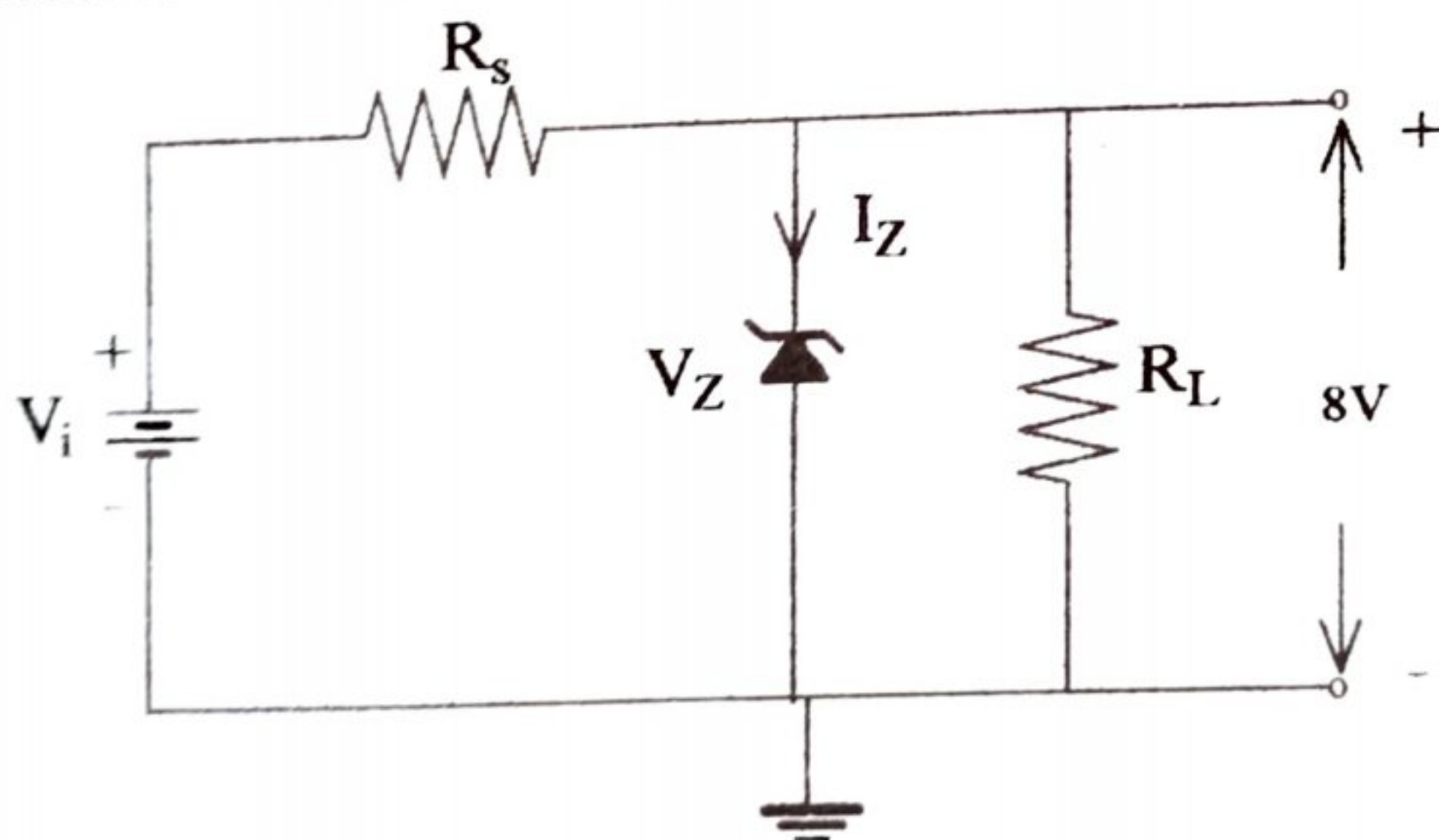
Max. Marks: 100

PART - A

10 x 2 = 20 Marks

Answer all the questions

1. Identify and define the breakdown due to increased thermal energy at the junction.
2. The Zener diode in the circuit shown in Fig.2 has a knee current of 3mA, $R_s = 600\Omega$, and an input voltage of 12V. Find the maximum load current when the output voltage is constant at 8V.



3. In an electronics company, as a graduate trainee, you are in a position to design an alarm circuit. Choose the relevant component to develop the alarm circuit for a door system in which the reverse

current increases when the junction is exposed to light. Explain the working principle of the component chosen.

4. Justify why CE configuration is preferred as an amplifier in BJT.
5. The AC gain of a transistor β is 150. What is the change in the collector current in the transistor whose base current changes by $130\mu\text{A}$?
6. Sketch the CC configuration using a PNP transistor.
7. The drain current of JFET varies from 3.2mA to 3.8mA , thereby changing the reverse gate voltage of JFET from 5.4V to 5V . Calculate the value of transconductance.
8. You are selecting two transistors of voltage-controlled type. The gate terminal of the first transistor is not insulated, and another transistor's gate terminal is insulated with SiO_2 . How do you distinguish between the two transistors?
9. A 2mA meter movement with an internal resistance of $200\ \Omega$ is to be converted into $(0-200)\ \text{mA}$. Compute the value of shunt resistance required.
10. Determine the unknown frequency of the input signal applied to the vertical deflection plate when the horizontal deflection plate frequency is 2kHz for the Lissajous pattern obtained as in fig(1).



fig(1)

PART - B

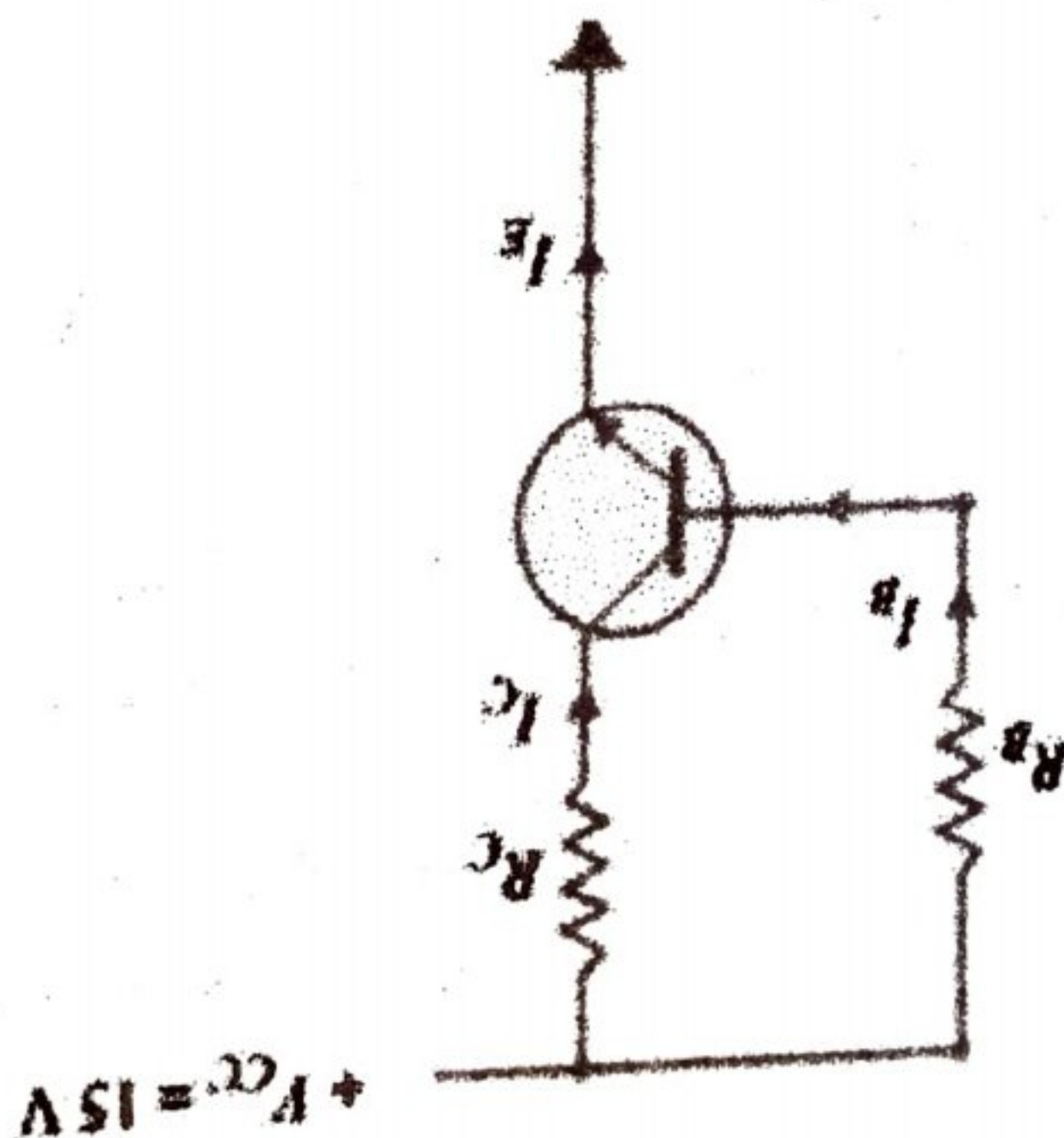
$4 \times 15 = 60$ Marks

Answer any Four Questions

11. a) Explain the operation of a full wave rectifier using two diodes with the help of input and output waveforms

(10)

b) Design the base resistor (R_B) for the circuit given in figure if the selected operating point is $V_{CE} = 9V$ and $I_C = 3mA$. You are supplied with a fixed $15V$ dc supply and a silicon transistor with $\beta = 100$. Take $V_{BE} = 0.7V$. Also, calculate the value of load resistance (R_C) that would be employed.



(5)

12. The input to a bridge rectifier is through a step-down transformer with a turns ratio of 10:1. The supply voltage is $230V$ at $60Hz$. The load resistance is 260Ω . The resistance of the secondary winding of the transformer is 5Ω , and the diode forward resistance is 2Ω . Determine (i). RMS voltage on the secondary side of the transformer. (ii). DC output voltage. (iii). DC output current. (iv). RMS value of the output current. (v). DC output power. (vi). AC input power. (vii). Rectification efficiency. (viii). Peak Inverse Voltage (PIV). (ix). Output frequency. (x). Ripple factor.

13. a) Explain the operation of a transistor in CE configuration with input and output characteristics. Also, explain the characteristics of the amplifier. (12)
- b) Explain the operation of a transistor as a switch. (03)
14. a) As an Electronics Engineer, identify a voltage control device, where the control voltage is a negative input. Explain its working principle, its characteristics curve, and its applications. (12)
- b) A gate current of 20nA is produced from a unipolar device when the gate is reverse-biased with 10V . Determine the resistance between the gate and the source. (3)
15. a) Sketch the basic construction of a normally OFF MOS transistor, which turns ON when the applied gate-source voltage V_{GS} is greater than or equal to the threshold voltage V_T of the device. Also, explain the characteristics of its working. (12)
- b). Also, a D-MOSFET has parameters of $I_{DSS} = 20\text{mA}$ and $V_{GS(off)} = -8\text{V}$
- (i) Is this device an n-channel or a p-channel?
- (ii) Calculate the drain current I_D at $V_{GS} = -4\text{V}$ (3)
16. Draw the block diagram of an oscilloscope and explain its operation in detail. Also, explain how the voltage measurement is performed using an oscilloscope.

PART – C

Answer the following

1 x 20 = 20 Marks

17. a). Explain the operation of the regulated power supply in detail with a neat block diagram. (10)
- b) With a suitable diagram, design a multi-range dc milli ammeter using a basic movement with an internal resistance $R_m = 50$ and a full-scale deflection current of 1mA . The ranges required are $0\text{-}10\text{mA}$, $0\text{-}125\text{mA}$, $0\text{-}250\text{mA}$, and $0\text{-}500\text{mA}$. (05)
- c) Explain how CMOS is used as an inverter. (05)
